



# Rule of Thumbs

- Use the spark-sql to test out individual SQL statement before use it in pyspark
- Make sure proper SQL functions such as max, min, desc are imported in pyspark before using them
- Pyspark codes should be the primary tools for developing your big data analytic applications
- Use temp view for SQL tables
- The following commands can be run under VM \$ **pyspark** or Databricks Community Edition

# Flight summary 2015 data

- #INPUT: dataframe
- #note change the third column name of flight 2015 from count #into fnum to avoid confusion

```
flightData2015 = spark\
```

```
.read\
```

```
.option("inferSchema", "true")\
```

```
.option("header", "true")\
```

```
.csv("file:///home/changs/2015-summary.csv")
```

#note if you run it in the windows 10 system the last line is

```
.csv("file:///c:/users/changs/downloads/2015-summary.csv")
```

# #create a temporary view as table

```
flightData2015.createOrReplaceTempView("flight_data_2015")
```

- What is the name of the view/table?
- Which one of the following SQL statement is correct in pyspark?
  1. Flightdata=spark.sql("""SELECT \* FROM flightData2015""")
  2. Flightdata=spark.sql("""SELECT \* FROM flight\_Data\_2015""")

# Run SQL statements in pyspark

```
sqlWay3=spark.sql("""  
SELECT ORIGIN_COUNTRY_NAME, DEST_COUNTRY_NAME, fnum  
FROM flight_data_2015  
""")  
# Note that you can also include the following statements before """)  
WHERE fnum > 300  
ORDER BY ORIGIN_COUNTRY_NAME, fnum DESC
```

# #Write the output as a csv file

```
sqlWay3.write.format("csv").mode("overwrite").option("sep",  
"\t").save("file:///home/changs/output3")
```

- Replace "\t" by ", " to generate a csv file
- # In Linux VM1 command line:

```
cat output3/part* > mycsvfile.tsv
```

```
cat output3/mycsvfile.tsv
```

- **This is one way to consolidate all output files in one!**

# Import max function

```
from pyspark.sql.functions import max
```

```
#compare the following statements
```

```
flightData2015.select(max("fnum")).take(1)
```

```
flightData2015.select(max("fnum")).show()
```

# Another Max operation

```
maxSql = spark.sql("""  
SELECT DEST_COUNTRY_NAME, sum(fnum) as destination_total  
FROM flight_data_2015  
GROUP BY DEST_COUNTRY_NAME  
ORDER BY destination_total DESC  
LIMIT 2  
""")
```

```
maxSql.show()
```



# Switching back to dataframe (ref. maxsql)

```
from pyspark.sql.functions import desc
```

```
flightData2015\  
  .groupBy("DEST_COUNTRY_NAME")\  
  .sum("fnum")\  
  .withColumnRenamed("sum(fnum)", "destination_total")\  
  .sort(desc("destination_total"))\  
  .limit(2)\  
  .show()
```

# Your Turn

- Build a notebook based on this lecture using Databricks Community Edition
- Export this notebook as part of your lecture notes.
- Make sure that you comments on each command

# Conclusion

- Two ways of manipulating and retrieving data in pyspark
  - Data frame
  - Call SQL statements
- It is the personal preference of which way one should manipulate and retrieve data. However, you should study dataframe way of programming because it has the similar style to the Spark RDD way for building a computational pipeline